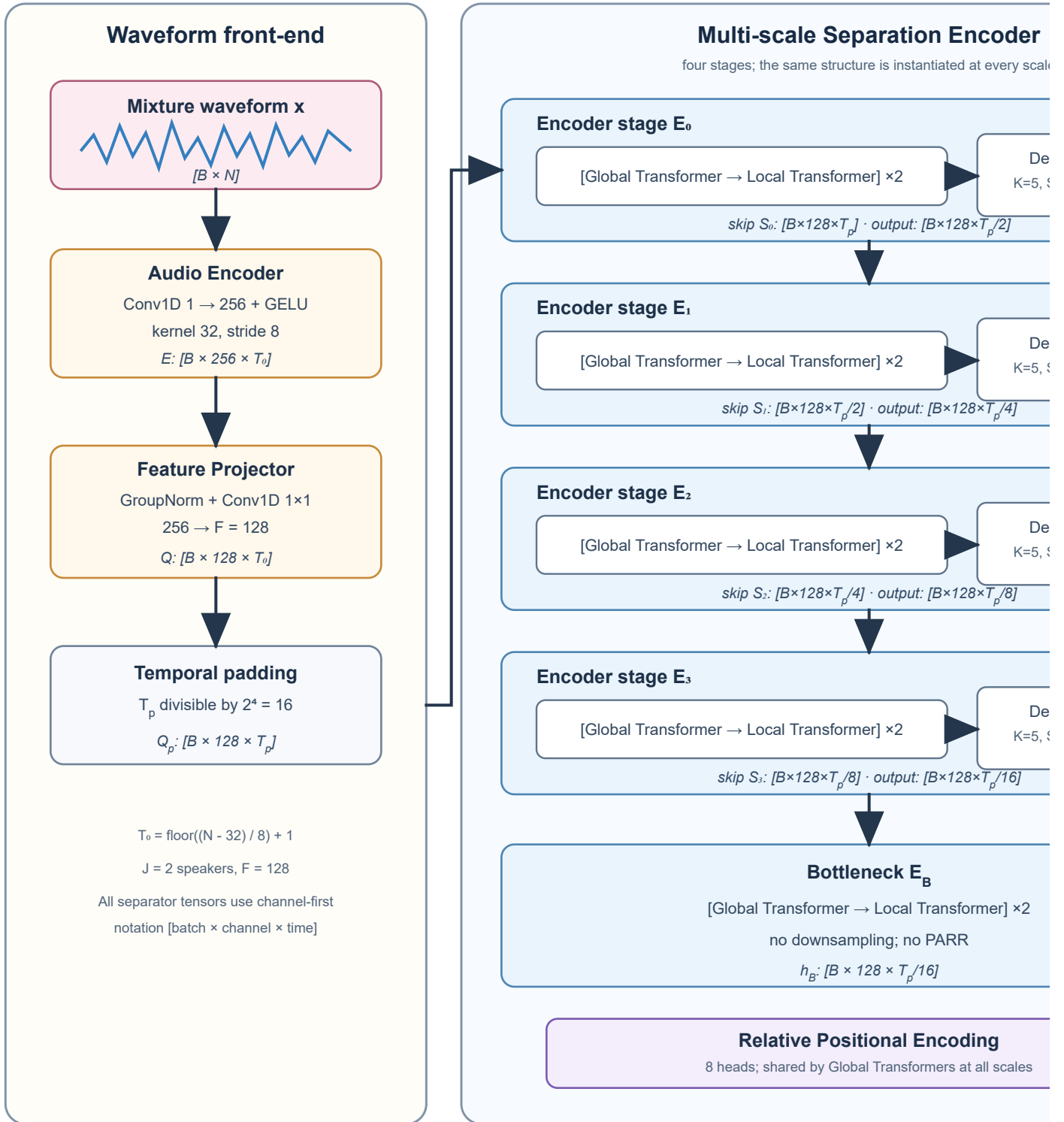


A. End-to-end framework



B. PARR exact forward path

The temporal core is shared across all stages and speaker branches; only Norm_r and γ_r are stage-specific.

